

A monotone Lipschitz scalar obstruction for threaded long-knife multidivisions

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Abstract

I verify and repair a proposed construction concerning the non-prime-power obstruction in the multilayered cake model of Igarashi and Meunier. The repair gives a rigorous monotone Lipschitz valuation counterexample for a natural restricted class of long-knife divisions, called here *threaded* long-knife divisions: every piece receives exactly one long-knife interval in every layer, and hence, under feasibility when the number of layers equals the number of pieces, each piece is an itinerary through all columns. For every non-prime-power q —in particular for $q = 6$ —there are q identical bounded monotone Lipschitz valuation functions on q layers such that no threaded complete feasible division obtained by $q - 1$ long knives is envy-free.

The result is therefore genuine partial progress toward the open question asking for valuation-function counterexamples for non-prime-power q . It is not, as written, a full solution to the broadest interpretation of that open question, because it does not rule out non-threaded long-knife divisions in which a piece may receive several consecutive intervals from one layer and no interval from another. The valuations constructed below are also not additive density valuations.

1 Status of the result

The paper of Igarashi and Meunier proves that, when q is a prime power and $m \leq q \leq n$, there exists an envy-free feasible contiguous multidivision obtainable with $q - 1$ long knives under closed preferences [2]. They point out that for non-prime-power q , examples of Avvakumov–Karasev and Panina–Živaljević give obstructions under the choice-function model, and they ask whether a counterexample exists for some natural valuation functions. The construction below converts the Avvakumov–Karasev pseudo-equivariant obstruction into scalar valuations, but only for the threaded/bijective long-knife submodel defined in Definition 2.1.

Thus the correct conclusion is:

- If “ $q - 1$ long knives” is formalized as the threaded/bijective family in which every piece receives one interval in every layer, then Theorem 7.1 gives a solution in the monotone Lipschitz valuation model.
- Under the broader formalization allowing a piece to receive multiple intervals from the same layer and no interval from another layer, Theorem 7.1 is partial progress only. Section 8 explains the obstruction to claiming more.

2 Model and the threaded long-knife family

Fix an integer $q \geq 2$. A q -layered piece is a tuple

$$L = (L_1, \dots, L_q), \quad L_\ell \subseteq [0, 1],$$

where each L_ℓ is a finite union of closed intervals. As usual, sets that differ by a null set are identified. Put

$$\mu(L) = \sum_{\ell=1}^q \lambda(L_\ell), \quad d(L, L') = \mu(L \Delta L') = \sum_{\ell=1}^q \lambda(L_\ell \Delta L'_\ell).$$

Let $\mathcal{X}_q$ denote the resulting metric space of q -layered pieces. A valuation is *Lipschitz* if it is Lipschitz for d , and *monotone* if

$$L_\ell \subseteq L'_\ell \quad (\ell = 1, \dots, q) \implies v(L) \leq v(L').$$

These are the monotonicity and Lipschitz conditions used by Igarashi and Meunier for valuation functions.

A complete feasible multidivision into q pieces is a q -tuple $A = (A_1, \dots, A_q)$ that partitions the q -layered cake up to endpoints and null sets, and such that each A_j is non-overlapping: for $\ell \neq \ell'$, the sets $(A_j)_\ell$ and $(A_j)_{\ell'}$ intersect in a null set. It is contiguous if every $(A_j)_\ell$ is an interval, possibly empty.

Let

$$\Delta^{q-1} = \{x = (x_1, \dots, x_q) \in \mathbb{R}_{\geq 0}^q : x_1 + \dots + x_q = 1\}.$$

For $x \in \Delta^{q-1}$ define

$$s_r(x) = x_1 + \dots + x_r, \quad I_r(x) = [s_{r-1}(x), s_r(x)] \quad (r = 1, \dots, q),$$

where $s_0(x) = 0$. Thus x is the vector of the lengths of the q columns cut by $q - 1$ long knives.

Definition 2.1 (Threaded long-knife multidivision). A complete feasible multidivision $A = (A_1, \dots, A_q)$ into q pieces is called a *threaded long-knife multidivision* if there are $x \in \Delta^{q-1}$ and permutations

$$\sigma_1, \dots, \sigma_q \in S_q$$

such that

$$(A_j)_\ell = I_{\sigma_j(\ell)}(x) \quad (j, \ell \in [q]), \tag{2.1}$$

up to null sets, and for each layer ℓ the map $j \mapsto \sigma_j(\ell)$ is a permutation of $[q]$.

The last condition says that in every layer the q long-knife intervals are assigned bijectively to the q pieces. Since each σ_j is itself a permutation, each single piece uses every positive column exactly once, across the layers; this is the itinerary or threading property.

Remark 2.2. The family in Definition 2.1 contains the balanced long-knife divisions encoded in the configuration-space construction of Igarashi and Meunier when $m = q$: in their construction the bundle names form a bijection in each layer, and the injectivity of the layer map forces each bundle to use distinct columns across distinct layers. Definition 2.1 is nevertheless not the broadest possible interpretation of “a division obtained by $q - 1$ long knives.”

3 The Avvakumov–Karasev input

Let

$$\mathbf{c} = (1/q, \dots, 1/q) \in \Delta^{q-1}.$$

We use the following direct corollary of Avvakumov and Karasev’s non-prime-power construction.

Theorem 3.1 (Avvakumov–Karasev corollary). *If q is not a prime power, then there is a continuous map*

$$F : \Delta^{q-1} \longrightarrow \Delta^{q-1} \setminus \{\mathbf{c}\}$$

with the following compatibility property. Let H, H' be two proper faces of Δ^{q-1} of the same dimension, and let π be a permutation of $[q]$ that sends the nonzero coordinate positions of H to the nonzero coordinate positions of H' preserving their order, while sending the zero coordinate positions of H bijectively to the zero coordinate positions of H' . Then

$$F(\pi x) = \pi F(x) \quad \text{for every } x \in H. \quad (3.1)$$

Here π acts on Δ^{q-1} by permuting coordinates.

Proof. Avvakumov and Karasev define precisely the pseudo-equivariance relations (3.1) for the segment partition problem with one sort of empty part. Their Theorem 4.3 gives, for every non-prime-power q , a pseudo-equivariant map

$$\partial\Delta^{q-1} \longrightarrow \Delta^{q-1}$$

whose image has zero linking number with the barycenter. As they explain immediately before that theorem, zero linking number is exactly the obstruction to extending this boundary map through the simplex while avoiding the barycenter; moreover the pseudo-equivariance requirements are imposed only on proper faces, hence an arbitrary extension over the interior introduces no new compatibility condition. The resulting extension is the required F . $\square$

The role of (3.1) is to make functions well-defined when zero-length intervals are moved, inserted, deleted, or relabelled. The following elementary consequence is the exact form needed later.

Lemma 3.2 (Coordinate compatibility). *Suppose (x, r) and (y, s) describe the same interval in the following sense: after deleting all zero-length coordinates of x and y , the ordered lists of positive lengths are the same, and the interval $I_r(x)$ corresponds to $I_s(y)$; if the interval has length zero, all zero intervals are regarded as the same empty part. Then*

$$F_r(x) = F_s(y).$$

Proof. If x has no zero coordinate, then $y = x$ and $r = s$. Otherwise x lies in a proper face. The stated relation between (x, r) and (y, s) is exactly implemented by one of the permutations π in Theorem 3.1, with $y = \pi x$ and $s = \pi(r)$. Equation (3.1) gives $F(y) = \pi F(x)$, so

$$F_s(y) = F_{\pi(r)}(\pi x) = F_r(x).$$

$\square$

4 A scalar potential on threaded pieces

For $x \in \Delta^{q-1}$ and $\sigma \in S_q$, define the layered itinerary

$$P(x, \sigma)_\ell = I_{\sigma(\ell)}(x) \quad (\ell = 1, \dots, q).$$

Let

$$\mathcal{Y}_q = \{P(x, \sigma) : x \in \Delta^{q-1}, \sigma \in S_q\} \subseteq \mathcal{X}_q.$$

Because $\Delta^{q-1} \times S_q$ is compact and the map $(x, \sigma) \mapsto P(x, \sigma)$ is continuous for the metric d , the itinerary space $\mathcal{Y}_q$ is compact.

Define

$$\phi_0(P(x, \sigma)) = F_{\sigma(1)}(x). \quad (4.1)$$

Lemma 4.1. *The formula (4.1) defines a continuous function $\phi_0 : \mathcal{Y}_q \rightarrow \mathbb{R}$.*

Proof. The only possible ambiguity in representing a point of $\mathcal{Y}_q$ by a pair (x, σ) comes from zero-length intervals. If

$$P(x, \sigma) = P(y, \tau)$$

in the metric quotient, then the layer-one intervals $I_{\sigma(1)}(x)$ and $I_{\tau(1)}(y)$ are the same interval after applying the zero-interval identifications described in Lemma 3.2. Therefore

$$F_{\sigma(1)}(x) = F_{\tau(1)}(y),$$

so (4.1) is well-defined.

For continuity, let $Q : \Delta^{q-1} \times S_q \rightarrow \mathcal{Y}_q$ be $Q(x, \sigma) = P(x, \sigma)$. The composition $\phi_0 \circ Q$ is the continuous function $(x, \sigma) \mapsto F_{\sigma(1)}(x)$. Since Q is a continuous surjection from a compact space to a Hausdorff space, it is a quotient map. Hence ϕ_0 is continuous. $\square$

Since F avoids the barycenter, the coordinate spread of $F(x)$ is uniformly positive:

$$\gamma := \min_{x \in \Delta^{q-1}} \left(\max_{r \in [q]} F_r(x) - \min_{r \in [q]} F_r(x) \right) > 0. \quad (4.2)$$

Indeed, the displayed spread is continuous, and inside Δ^{q-1} it vanishes exactly at $\mathbf{c}$.

Lemma 4.2 (Separation on threaded divisions). *Let $A = (A_1, \dots, A_q)$ be a threaded long-knife multidivision with cut-length vector x . Then each A_j lies in $\mathcal{Y}_q$, and*

$$\max_j \phi_0(A_j) - \min_j \phi_0(A_j) \geq \gamma.$$

Proof. By definition, $A_j = P(x, \sigma_j)$ for permutations $\sigma_j \in S_q$. The condition that the layer-one intervals are assigned bijectively says that

$$\{\sigma_1(1), \dots, \sigma_q(1)\} = [q].$$

Therefore the multiset of potential values is

$$\{\phi_0(A_j) : j \in [q]\} = \{F_1(x), \dots, F_q(x)\}.$$

The claimed lower bound is exactly (4.2). $\square$

5 Lipschitz approximation and monotone extension

The potential ϕ_0 is continuous but need not be Lipschitz. We first replace it by a uniformly close Lipschitz potential on $\mathcal{Y}_q$.

Lemma 5.1 (Lipschitz approximation on compact metric spaces). *Let M be a compact metric space and let $f : M \rightarrow \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there is a Lipschitz function $g : M \rightarrow \mathbb{R}$ such that $\|f - g\|_\infty < \varepsilon$.*

Proof. Choose $\delta > 0$ such that $d(u, v) < \delta$ implies $|f(u) - f(v)| < \varepsilon$. Let $\omega = \max f - \min f$. Choose $C > 0$ with $C\delta > \omega + \varepsilon$. Define

$$g(u) = \inf_{v \in M} (f(v) + Cd(u, v)).$$

The function g is C -Lipschitz. Since the term $v = u$ is allowed, $g(u) \leq f(u)$. If $d(u, v) < \delta$, then $f(v) + Cd(u, v) \geq f(u) - \varepsilon$. If $d(u, v) \geq \delta$, then

$$f(v) + Cd(u, v) \geq \min f + C\delta > \max f + \varepsilon \geq f(u) - \varepsilon.$$

Thus $g(u) > f(u) - \varepsilon$ for every u , and hence $\|f - g\|_\infty < \varepsilon$. $\square$

Apply Lemma 5.1 to ϕ_0 on $\mathcal{Y}_q$ with $\varepsilon = \gamma/4$. Choose a Lipschitz function

$$\phi_1 : \mathcal{Y}_q \rightarrow \mathbb{R} \quad \text{such that} \quad \|\phi_1 - \phi_0\|_\infty < \gamma/4. \quad (5.1)$$

Let K be a Lipschitz constant for ϕ_1 . Extend ϕ_1 to all layered pieces by the McShane formula

$$\Phi(L) = \inf_{Y \in \mathcal{Y}_q} (\phi_1(Y) + Kd(L, Y)). \quad (5.2)$$

Then $\Phi : \mathcal{X}_q \rightarrow \mathbb{R}$ is K -Lipschitz and agrees with ϕ_1 on $\mathcal{Y}_q$.

Combining Lemma 4.2 and (5.1), every threaded long-knife multidivision $A = (A_1, \dots, A_q)$ satisfies

$$\max_j \Phi(A_j) - \min_j \Phi(A_j) = \max_j \phi_1(A_j) - \min_j \phi_1(A_j) \geq \gamma/2. \quad (5.3)$$

Now set

$$v(L) = \Phi(L) + (K + 1)\mu(L). \quad (5.4)$$

Lemma 5.2. *The function $v : \mathcal{X}_q \rightarrow \mathbb{R}$ is bounded, Lipschitz, and monotone. After adding a constant, it may be made nonnegative without changing any preference comparison.*

Proof. The space $\mathcal{X}_q$ has diameter at most q because each layer has length at most 1. The function Φ is Lipschitz on this bounded space, hence bounded; also $0 \leq \mu(L) \leq q$. Thus v is bounded. It is Lipschitz because Φ is K -Lipschitz and $L \mapsto \mu(L)$ is 1-Lipschitz for d .

For monotonicity, suppose $L_\ell \subseteq L'_\ell$ for every layer. Then

$$d(L, L') = \mu(L') - \mu(L).$$

Therefore

$$\begin{aligned} v(L') - v(L) &= \Phi(L') - \Phi(L) + (K + 1)(\mu(L') - \mu(L)) \\ &\geq -Kd(L, L') + (K + 1)d(L, L') \\ &= d(L, L') \geq 0. \end{aligned}$$

Adding a constant clearly preserves all stated properties except the numerical lower bound, and it preserves all preference comparisons. $\square$

6 Equal total length in feasible q -by- q divisions

The reason the monotone length term does not affect the obstruction is the following elementary fact.

Lemma 6.1. *In every complete feasible multidivision of a q -layered cake into q pieces, each piece has total length 1.*

Proof. For each piece A_j and almost every $t \in [0, 1]$, let

$$N_j(t) = \#\{\ell \in [q] : t \in (A_j)_\ell\}.$$

Completeness gives

$$\sum_{j=1}^q N_j(t) = q$$

for almost every t , since all q layer-points above t are allocated. Feasibility gives $N_j(t) \leq 1$ for every j and almost every t , since a single feasible piece cannot contain two different layers at the same horizontal coordinate except on a null set. With q nonnegative integers bounded above by 1 and summing to q , all of them must equal 1. Hence $N_j(t) = 1$ for every j and almost every t . Integrating gives

$$\mu(A_j) = \int_0^1 N_j(t) dt = 1.$$

□

Consequently, for every threaded complete feasible multidivision $A = (A_1, \dots, A_q)$,

$$v(A_j) - v(A_k) = \Phi(A_j) - \Phi(A_k) \quad (j, k \in [q]),$$

and (5.3) gives

$$\max_j v(A_j) - \min_j v(A_j) \geq \gamma/2. \quad (6.1)$$

7 Main theorem

Theorem 7.1. *Let q be not a prime power. There is an instance with $m = n = q$ in which all q agents have the same bounded monotone Lipschitz valuation function v and in which no threaded complete feasible multidivision obtained by $q - 1$ long knives is envy-free. In particular, this holds for the smallest non-prime-power value $q = 6$.*

Proof. Let all agents have the valuation v defined in (5.4). Suppose that a threaded complete feasible multidivision

$$A = (A_1, \dots, A_q)$$

were envy-free. Since $n = q$, a surjective assignment of agents to pieces is a bijection. Since all agents have identical valuations, an assigned piece is acceptable to its assigned agent only if it is a maximizer of v among $A_1, \dots, A_q$. A bijective envy-free assignment would therefore require every piece to be a maximizer, hence

$$v(A_1) = \dots = v(A_q).$$

This contradicts (6.1). Thus no threaded complete feasible long-knife multidivision is envy-free. □

8 Why this is not the full broad long-knife problem

The proof of Theorem 7.1 uses the threaded hypothesis in one essential place: Lemma 4.2. There we need the layer-one intervals assigned to the q pieces to have labels $1, \dots, q$ exactly once, and we need each individual piece to encode the whole cut-length vector by using every column once across the layers.

If a “long-knife division” is allowed to assign the q intervals of a layer non-bijectively to the q pieces, this encoding fails. For example, take the same cut vector in all layers and assign the entire ℓ -th layer to the ℓ -th piece, for $\ell = 1, \dots, q$. This gives a complete feasible contiguous division if one allows a piece to receive several adjacent long-knife intervals from one layer and no intervals from other layers. The pieces are whole layers, not itineraries $P(x, \sigma)$, and a single piece no longer contains enough information to recover the vector x of all long-knife lengths. Consequently the scalar function (4.1) is not defined on those pieces in a way that would reproduce the Avvakumov–Karasev coordinate spread.

This is the exact point at which the initial proposed argument overclaimed. The argument is rigorous for threaded/bijective long-knife divisions, and it shows that the choice-function obstruction can be made into bounded monotone Lipschitz scalar valuations on that configuration space. It does not yet prove that every broader long-knife multidivision is obstructed. Nor does it produce additive valuations of the form

$$L \mapsto \sum_{\ell=1}^q \int_{L_\ell} w_\ell(t) dt.$$

The valuation v is deliberately shape-dependent: it reads the itinerary of a piece through the long-knife columns.

References

- [1] S. Avvakumov and R. Karasev, *Envy-free division using mapping degree*, *Mathematika* 67 (2021), no. 1, 36–53. Preprint: arXiv:1907.11183.
- [2] A. Igarashi and F. Meunier, *Envy-Free Division of Multilayered Cakes*, *Mathematics of Operations Research* 50 (2025), no. 3, 2261–2286. Preprint: arXiv:2106.02262.
- [3] G. Panina and R. Živaljević, *Envy-free division via configuration spaces*, arXiv:2102.06886.